

Proportion Inference

Modeling a count of successes

Download the section 13.Rmd handout to
STAT240/lecture/sect13-single-proportion.

Download the file chimpanzee.csv to
STAT240/data.

Modeling two numeric variables:

- Use a linear regression model
- β_1 is the unknown parameter of interest

Now, we will model a binary outcome:

- Based on the binomial
- p is the unknown parameter of interest

The chimpanzee data is based on an Emory University experiment.

- Chimpanzees choose from colored tokens
- One color is **selfish**, the other is **prosocial**
- Different combinations were studied

p is the probability (proportion) of prosocial. More specifically,

- Is $p_{partner}$ greater than 0.5?
- Is $p_{partner}$ the same as $p_{nopartner}$?

Let X be the number of prosocial choices out of n trials. We have model

$$X \sim \text{Binom}(n, p)$$

which relies on the BINS assumptions. n is known and p is the parameter of interest.

Consider chimpanzee A.

They made 60 prosocial choices out of 90 with a partner. We have $x_{obs} = 60$, giving point estimate

$$\hat{p} = \frac{x_{obs}}{n} = \frac{60}{90}$$

What is the error in this estimate? Let's try a simulation.

Recall that the normal is a good approximation to the binomial when $np(1 - p)$ is large.

$$X \dot{\sim} N(np, \sqrt{np(1 - p)})$$

$$\hat{p} \dot{\sim} N\left(p, \sqrt{\frac{p(1 - p)}{n}}\right)$$

$$\hat{p} \dot{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

- $E(\hat{p}) = p$
- Error decreases with n
- Error is a function of p

Inference for p is based on the binomial and normal.

- CI critical values
- Null distribution for a hypothesis test

This depends on n and p . Try simulating the probability for chimpanzee D, with no partner.

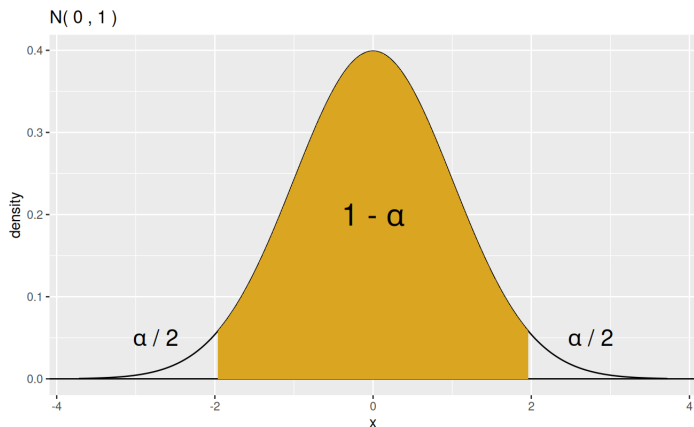
A CI, in general, is

point estimate $\pm$ critical value $\times$ standard error

- For p , the point estimate is $\hat{p}$
- The standard error is $\sqrt{\frac{p(1-p)}{n}}$

The critical value is from $N(0, 1)$.

Find a specific quantile with `pnorm`.



For a 90% CI, use 1.645.

Problem: can't find $\sqrt{\frac{p(1-p)}{n}}$. Use $\hat{p}$ instead:

$$\widehat{se}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

This is the **Wald** adjustment.

Wald CI for p :

$$\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

We are 90% confident that $p_{A,partner}$ is within (0.585, 0.748).

Another adjustment: add 2 successes and failures to “stabilize” data.

$$n_{AC} = n + 4, \quad \hat{p}_{AC} = \frac{X + 2}{n + 4}$$

$$\hat{se}(\hat{p}_{AC}) = \sqrt{\frac{\hat{p}_{AC}(1 - \hat{p}_{AC})}{n_{AC}}}$$

This is the **Agresti-Coull** adjustment.

A-C CI for p :

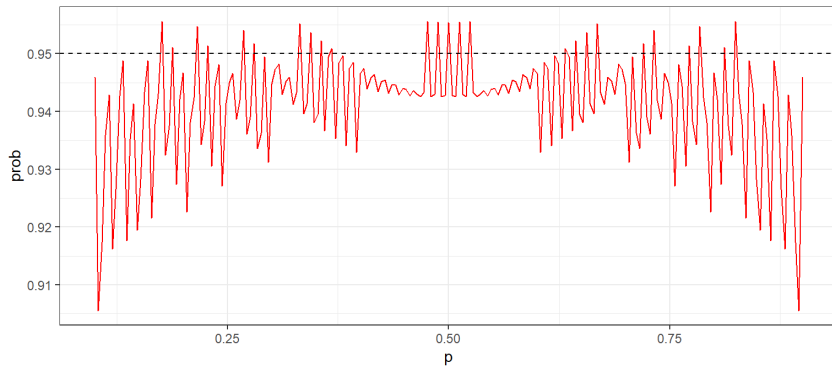
$$\hat{p}_{AC} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}_{AC}(1 - \hat{p}_{AC})}{n_{AC}}}$$

We are 90% confident that $p_{A,partner}$ is within (0.579, 0.74).

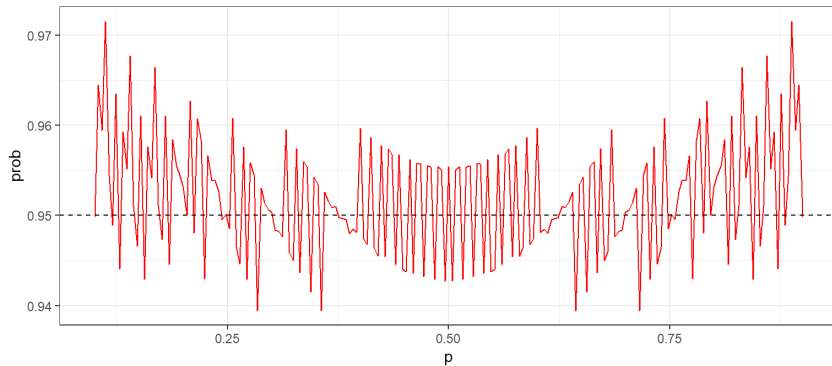
AC is usually preferable to Wald.

- Both methods are approximate
- AC tends to lead to intervals with coverage probability greater than $1 - \alpha$.

Wald Method Capture Probability

 $n = 90$ 

Agresti-Coull Method Capture Probability

 $n = 90$ 

We model the number of prosocial choices made by chimpanzee A (with a partner) as

$$\textit{Binom}(90, p)$$

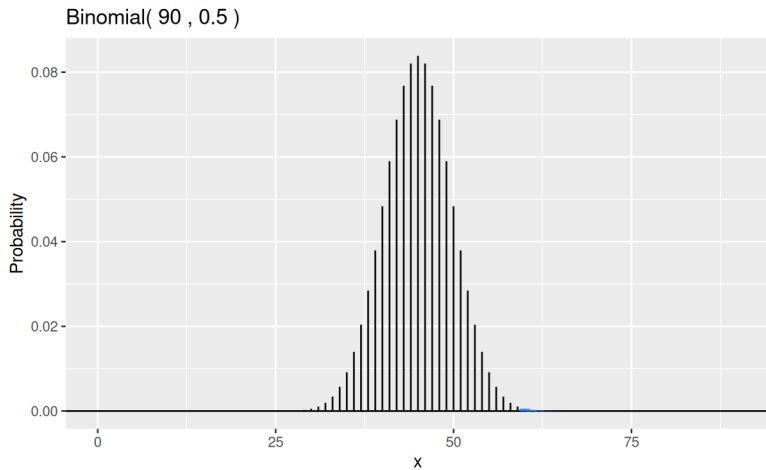
according to the BINS assumptions. Is chimpanzee A more prosocial than selfish?

$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p > 0.5$$

Our test statistic is X (count of prosocial choices) rather than $\hat{p}$. This gives null distribution

$$\text{Binom}(90, 0.5)$$

and an observed test statistic $x_{obs} = 60$. Values of X higher than 60 give more evidence for H_A .



The p-value is the probability above (and including) $x_{obs} = 60$ on our null distribution.

This is 0.001, which is strong evidence against the null. Chimpanzee A appears to be prosocial more than half the time.

The average prosocial rate with a partner is 0.59.
Chimpanzee F has a rate of 47/90.

Is this significantly *less* than 0.59?

- Set up hypotheses
- Identify null distribution
- Use test statistic $x_{obs} = 47$
- Calculate p-value in the *lower* direction

A psychic claims to be able to guess the suit of a random card without looking. 200 cards were drawn, and they guessed correctly 57 times.

Model:

$$X \sim \text{Binom}(200, p)$$

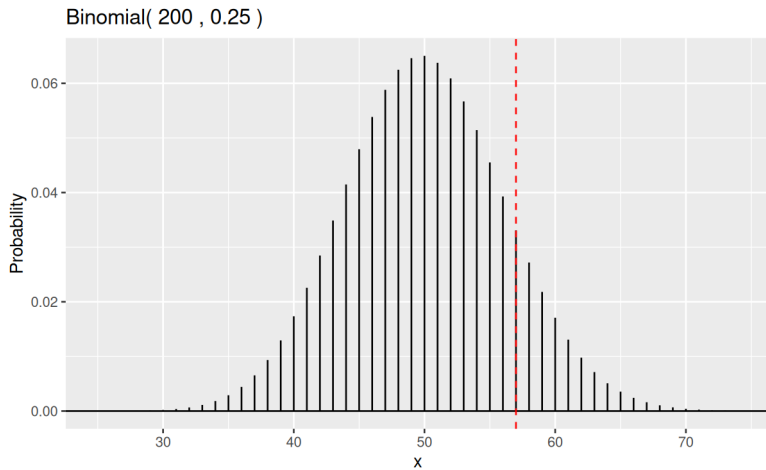
Our observed test statistic is $x_{obs} = 57$. Let's complete this test with $\alpha = 0.05$.

The problem suggests one-sided hypotheses:

$$H_0 : p = 0.25 \quad \text{versus} \quad H_A : p > 0.25$$

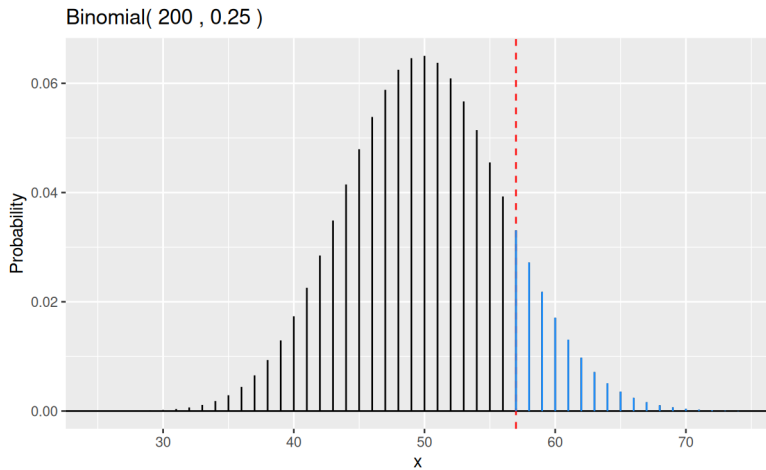
Under the null distribution, which outcomes are less likely than 57?

The distribution is not symmetric.



The outcomes as or less likely than x_{obs} are $X \leq 42$ and $X \geq 57$.

For $H_A : p > 0.25$, only $X \geq 57$ is relevant. We get a p-value of 0.145 and fail to reject H_0 .

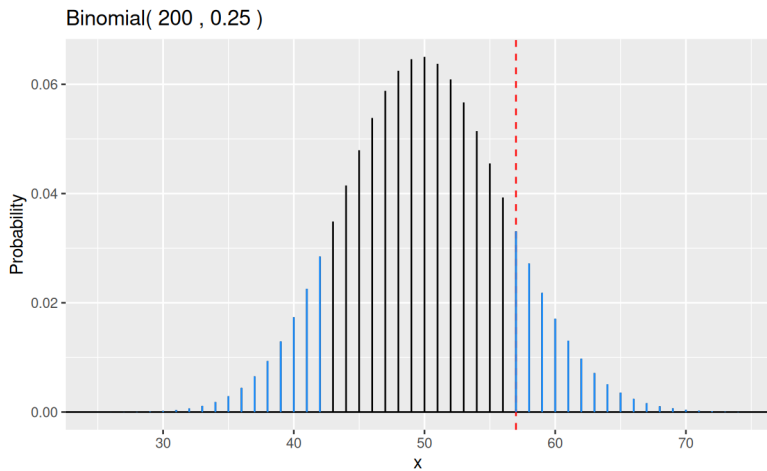


What if we had two-sided hypotheses?

$$H_0 : p = 0.25 \quad \text{versus} \quad H_A : p \neq 0.25$$

Now, we have to look for p being different from 0.25 in *both* directions.

Our p-value is $P(X \leq 42) + P(X \geq 57) = 0.253$.



In general, for different hypothesis directions:

- Test statistic does not change
- Null distribution does not change
- p-value changes
- Conclusions change

An alternative to the binomial test is a **Z test**. If the null hypothesis were true,

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

$$\hat{p} \sim N\left(0.25, \sqrt{\frac{0.25(0.75)}{200}}\right)$$

By standardization,

$$Z = \frac{\hat{p} - 0.25}{\sqrt{\frac{0.25(0.75)}{200}}} \dot{\sim} N(0, 1)$$

Our test statistic is the observed Z . and our null is $N(0, 1)$.

$$H_0 : p = 0.25 \quad \text{versus} \quad H_A : p \neq 0.25$$

$$z_{obs} = \frac{\frac{57}{200} - 0.25}{\sqrt{\frac{0.25(0.75)}{200}}} = 1.143$$

Is 1.143 consistent with $N(0, 1)$? For a two-sided test, we need the area outside of 1.143 in *both tails*.

$$P(Z \leq -1.143) + P(Z \geq 1.143)$$

The p-value is 0.253, just like the exact test.

